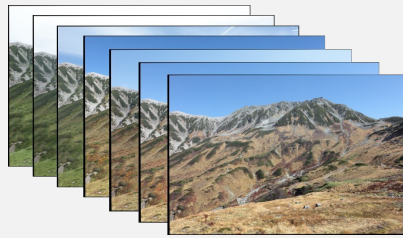
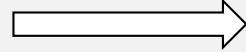


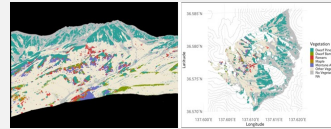
(a) Time-lapse imagery
(2011–2021)



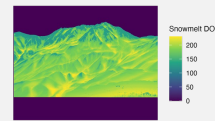
align,
classify,
geo-correct



Vegetation maps (2012, 2021)



Snowmelt DOY (2011–2021)



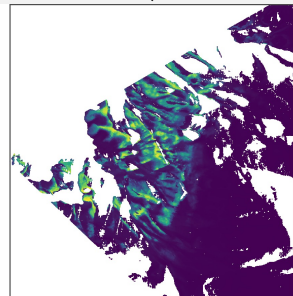
Terrain (DEM)

elevation, slope, TPI, TWI, aspect

(b) 2021 presence + environment



Model A: habitat suitability
environmental potential (seed layer)

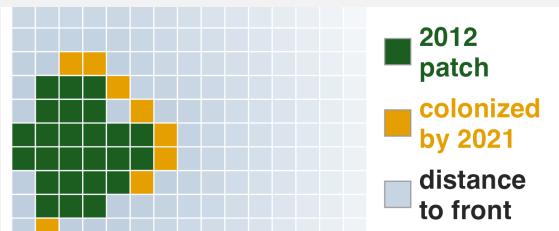


Habitat suitability surface

(c) 2012→2021 transitions
+ distance to front
+ environment



Model B: establishment
front distance + environment
(rhizome layer)



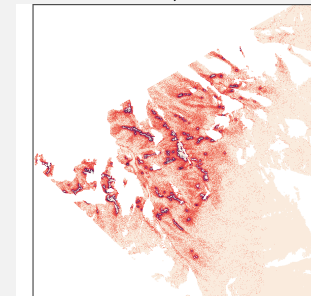
Establishment hugs the 2012 front

(d) **Model B applied annually**

Snowmelt scenarios
 $s = 0, -0.71, -2.24 \text{ d yr}^{-1}$



Projection: cellular automaton
annual establishment trial, 2022–2030



P(colonized by 2030), $s = 0$